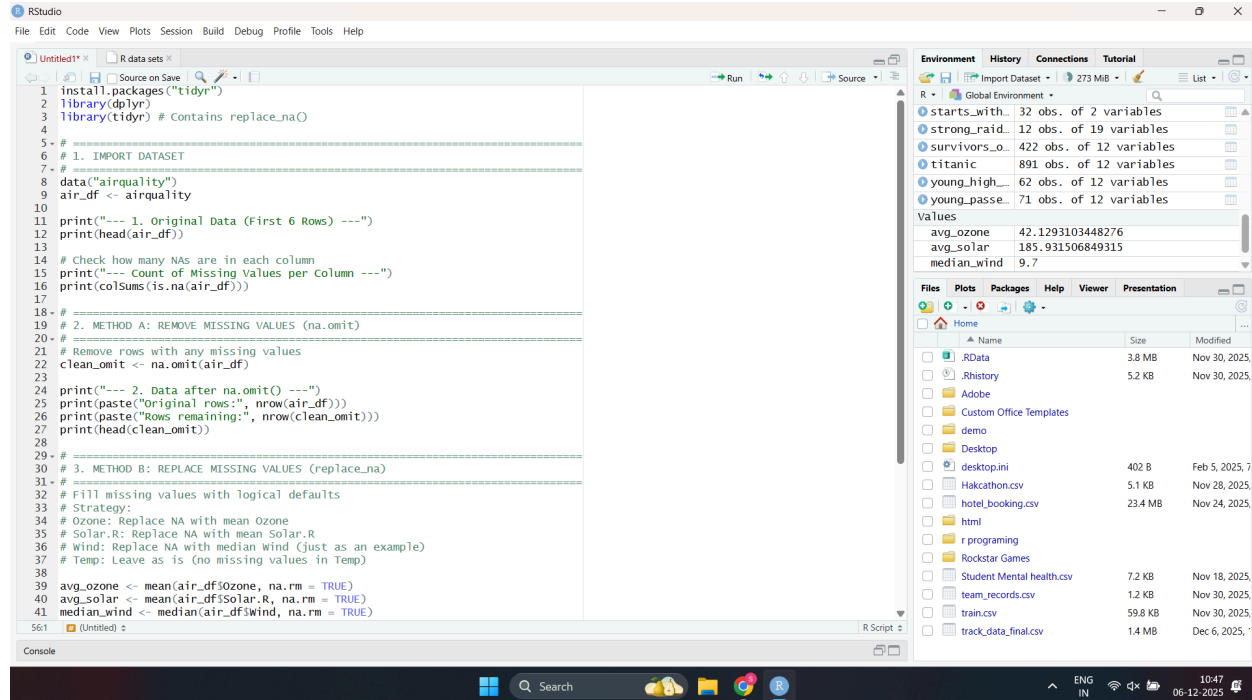


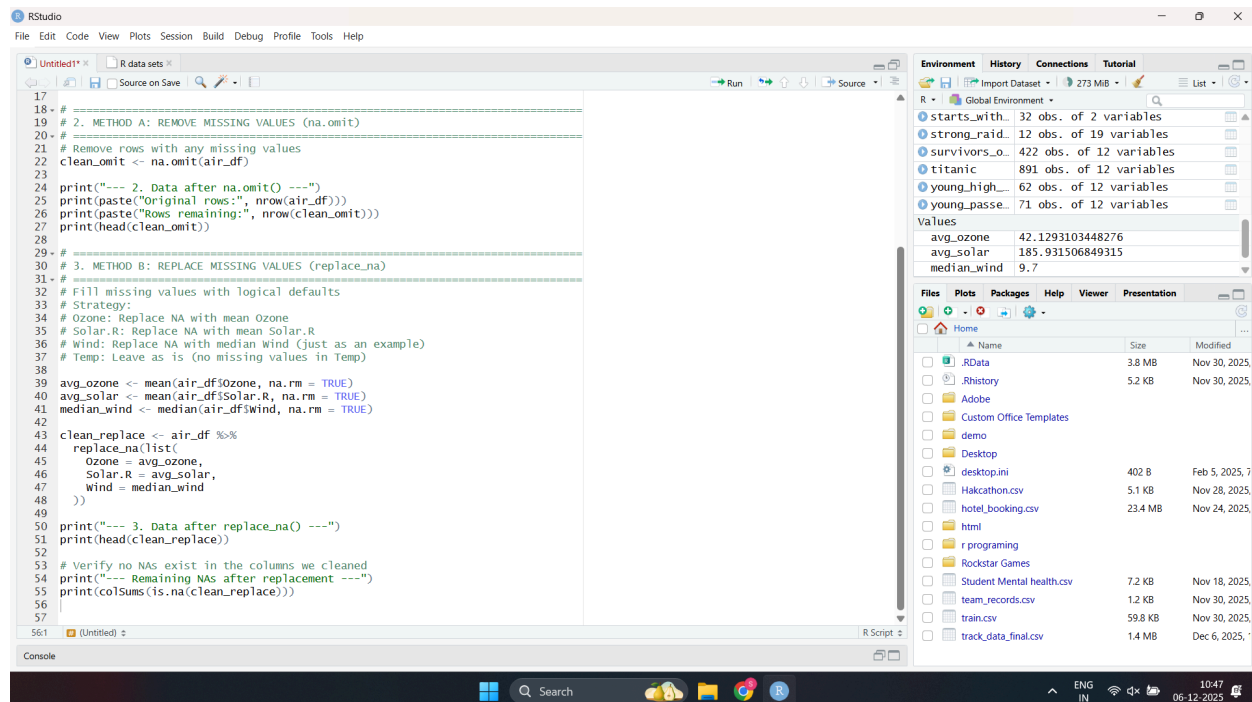
SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis SAS / SPSS / R

Aim - Applying basic data cleaning functions: handling missing values using `na.omit()/replace_na()` in R. import dataset.

Input :



```
1 install.packages("tidyr")
2 library(dplyr)
3 library(tidyr) # Contains replace_na()
4
5 # =====
6 # 1. IMPORT DATASET
7 # =====
8 data("airquality")
9 air_df <- airquality
10
11 print("--- 1. Original Data (First 6 Rows) ---")
12 print(head(air_df))
13
14 # Check how many NAs are in each column
15 print("--- Count of Missing Values per Column ---")
16 print(colSums(is.na(air_df)))
17
18 # =====
19 # 2. METHOD A: REMOVE MISSING VALUES (na.omit)
20 # =====
21 # Remove rows with any missing values
22 clean_omit <- na.omit(air_df)
23
24 print("--- 2. Data after na.omit() ---")
25 print(paste("Original rows:", nrow(air_df)))
26 print(paste("Rows remaining:", nrow(clean_omit)))
27 print(head(clean_omit))
28
29 # =====
30 # 3. METHOD B: REPLACE MISSING VALUES (replace_na)
31 # =====
32 # Fill missing values with logical defaults
33 # Strategy:
34 # Ozone: Replace NA with mean Ozone
35 # Solar.R: Replace NA with mean Solar.R
36 # Wind: Replace NA with median Wind (just as an example)
37 # Temp: Leave as is (no missing values in Temp)
38
39 avg_ozone <- mean(air_df$ozone, na.rm = TRUE)
40 avg_solar <- mean(air_df$solar.R, na.rm = TRUE)
41 median_wind <- median(air_df$wind, na.rm = TRUE)
42
43 clean_replace <- air_df %>%
44   replace_na(list(
45     ozone = avg_ozone,
46     solar.R = avg_solar,
47     wind = median_wind
48   ))
49
50 print("--- 3. Data after replace_na() ---")
51 print(head(clean_replace))
52
53 # Verify no NAs exist in the columns we cleaned
54 print("--- Remaining NAs after replacement ---")
55 print(colSums(is.na(clean_replace)))
56
57
```



```
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Output :